

R1 feeder geometry candidate — unselected

This is a separate geometry experiment, not a change to the evaluated Gen5 inputs or an accepted feeder. It retains the eight Gen5 source parts except the enclosure and cassettes, which are modeled as new STEP parts. The 3U proxies translate from six side cells into a central vertical lift corridor, then descend to the first release height. The supports are conceptual guides; actuation and launch retention are not designed.

External width grows from 530 to 570 mm, giving 9.5 mm cassette-to-inner-skin clearance and 5.0 mm cassette-to-track clearance per side. Exact static track/cassette overlap is $\{ '1': 0.0, '-1': 0.0 \}$ mm³. The simple aluminium-skin mass increment is 0.99 kg, not a revised installed-system mass. The new enclosure also changes host accommodation.

Sampled routes clear: 12 / 12. Each axis-aligned horizontal and vertical translation also clears a conservative full $340.5 \times 100 \times 100$ mm swept box against the listed fixed parts. This does not include tolerances, the moving lift/gripper, dynamics, cycle life, safety or launch loads.

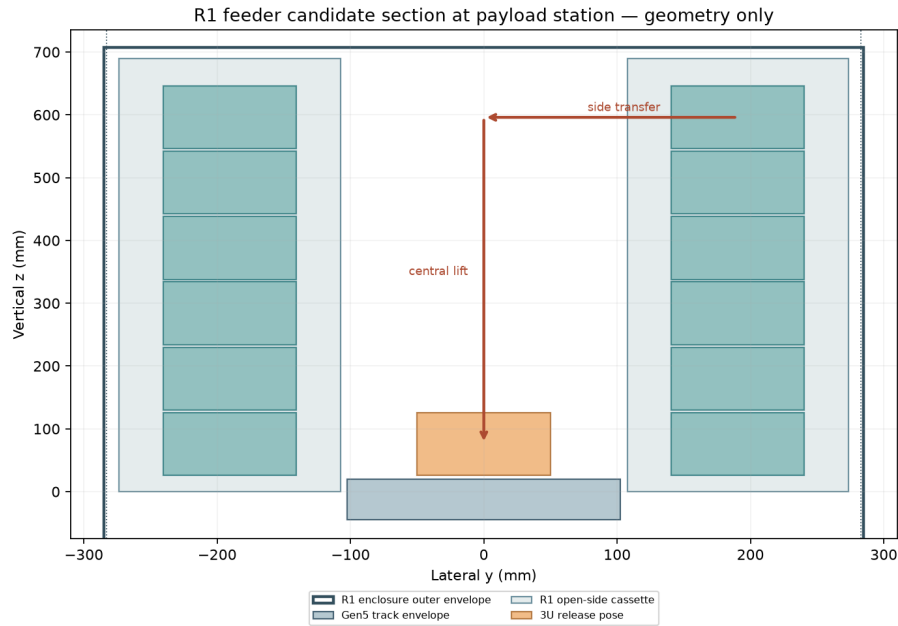


Figure 1: R1 geometry section

This is a parameter-derived section, not a CAD GUI or solver screenshot. The native FreeCAD review document is `cad/native/Feeder_Candidate_R1.FCStd`; its 21 named instances and FreeCAD-exported assembly STEP have read-back and interference results in `cad/FREECAD_FEEDER_R1.json`. Imported B-reps

have no native feature histories.

Open design gates

- no actuator, lift carriage or motor package
- no launch retention gate or load path
- no fastener/harness/thermal/tolerance clearance
- no provider ICD or named payload
- larger enclosure invalidates Gen5 mass, structural and host envelopes
- rigid swept-envelope screen excludes dynamics, elastic deflection and fault cases

The STEP part and assembly exports, hashes and every collision result are in `cad/FEEDER_CANDIDATE_R1.json`. The source is `cad/build_feeder_candidate_r1.py`.